

# Feldman–Hájek, but Non-Gaussian: Geometry of the Weighted Fourier Jet

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## Abstract

Feldman–Hájek theory gives an exact measure-class classification for Gaussian measures. What replaces it for a non-Gaussian product law under arbitrary linear mixing? We give a complete answer and it is rigid. In the Gaussian case, orthogonal mixing preserves the product structure. Outside the Gaussian case, this continuous symmetry disappears. Only signed permutations remain. We prove this by a weighted Fourier-jet argument.

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# 1 Introduction and Main Results

## 1.1 Historical context: from Kakutani to dense linear mixing

Kakutani’s theorem turns equivalence of infinite products into a scalar summability problem. The test is the product of one-coordinate Hellinger affinities. This product is positive exactly on the equivalence branch, and zero exactly on the singular branch [9]. This is one of the cleanest infinite-dimensional zero–one phenomena.

Its direct scope is coordinatewise. Sharp criteria for sequences of Euclidean and, more generally, Lie-group perturbations retain this product structure. These criteria identify Fisher information as the local Hellinger metric [3, 12]. If a bounded operator mixes every output coordinate with infinitely many inputs, the transformed coordinates cease to be independent and the affinity no longer tensorizes.

The Gaussian law conceals this obstruction. For the standard Gaussian product, every orthogonal operator on  $\ell^2$  has a measurable row-series realization and preserves the product law. Translation and linear measure-class changes are governed by the Cameron–Martin and Feldman–Hájek theories. Explicit Gaussian density criteria were developed by Shepp. Ramer’s theorem gives a nonlinear extension [1, 2, 4, 6, 10, 11]. A non-Gaussian product behaves differently. Exact independence rigidifies the orthogonal group to coordinate permutations and to those coordinate signs already preserving the one-dimensional law. The present paper identifies the corresponding measure-class enlargement of that discrete stabilizer. Non-Gaussian quasi-invariant measures on Hilbert and linear topological spaces have long been known [5, 13]; the issue here is instead a complete classification for dense linear mixing of one fixed i.i.d. product law.

There is a second obstacle. Every fixed finite output marginal can have a strictly positive density even when the full transformed law is singular to the reference product. Hence neither convergence on cylinder functions nor fixed-dimensional calculations detect the boundary. The correct completion must retain the entire directed family of projected likelihoods. Our Hellinger energy does precisely this: bounded projected energy is equivalent to positive global affinity, and a translation-ergodic zero–one law then upgrades positive affinity to equivalence.

## 1.2 The phase transition: continuous Gaussian symmetry and discrete rigidity

Let  $X$  be centered with finite first moment and characteristic function  $\phi(u) = \mathbb{E}e^{iuX}$ . The fundamental detector is

$$R_\mu(s, t) = t\phi'(s)\phi(t) - s\phi(s)\phi'(t). \quad (1)$$

It is the first variation of a bounded, centered two-frequency feature under a planar rotation. The correct geometric curve is the weighted Fourier jet

$$\Gamma_\mu(u) = (u\phi(u), \phi'(u)) \in \mathbb{C}^2. \quad (2)$$

Up to sign,  $R_\mu(s, t)$  is the wedge product of  $\Gamma_\mu(s)$  and  $\Gamma_\mu(t)$ . Hence zero defect means that the weighted jet lies in one fixed complex line through the origin. The resulting global linear ODE

has only Gaussian characteristic functions as solutions. This formulation is important. The unweighted curve  $u \mapsto (\phi(u), \phi'(u))$  is not a straight line even in the Gaussian case.

For a non-Gaussian coordinate law, a finite set of frequencies therefore detects every infinitesimal matrix direction. The corresponding Hellinger orbit is uniformly coercive in Frobenius norm near the discrete stabilizer, independently of dimension. In infinite dimensions the surviving continuous thickness is exactly Hilbert–Schmidt:

$$\text{exact stabilizer } \mathcal{P}_f \rightsquigarrow \text{measure-class stabilizer } \{T : TQ^{-1} - I \in \mathcal{S}_2 \text{ for some } Q \in \mathcal{P}_f\}.$$

This is the non-Gaussian phase transition studied below.

### 1.3 Main theorem: equivalence, energy, and the stabilizer

We now state the assumptions used for the full arbitrary-mixing theorem. Let  $f$  be a probability density on  $\mathbb{R}$ , put  $g = \sqrt{f}$ , and assume

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g &\in W^{1,2}(\mathbb{R}), \quad xg' \in L^2(\mathbb{R}), \\ \mathbb{E}_f X &= 0, \quad \mathbb{E}_f X^2 = 1, \quad f \neq (2\pi)^{-1/2} e^{-x^2/2}. \end{aligned} \tag{3}$$

All derivatives are weak derivatives. Cusps are allowed. No moment of order greater than two is assumed. The two Sobolev conditions are finite location and scale Fisher information.

Put  $H = \ell^2(\mathbb{N}; \mathbb{R})$  and  $\mu = f^{\otimes \mathbb{N}}$  on the product Borel space  $\mathbb{R}^{\mathbb{N}}$ . The space  $H$  indexes coefficients. A sample from  $\mu$  is not asserted to belong to  $H$ . Define

$$\begin{aligned} \Sigma_f &= \{\varepsilon \in \{-1, 1\} : \text{Law}(\varepsilon X) = \text{Law}(X)\}, \\ \mathcal{P}_f &= \{Q : Q \text{ is a coordinate permutation with signs in } \Sigma_f\}. \end{aligned} \tag{4}$$

Thus  $\mathcal{P}_f$  is the full signed-permutation group when  $f$  is even. It is the ordinary permutation group when  $f$  is not. For  $T \in \text{GL}(H)$ , let  $\Phi_T$  be the measurable row-series realization constructed in [Theorem 4.1](#). Write  $\nu_T = (\Phi_T)_\# \mu$ .

For probability measures  $\lambda, \nu$ , set

$$A(\lambda, \nu) = \int \sqrt{\frac{d\lambda}{dm} \frac{d\nu}{dm}} dm, \quad h(\lambda, \nu)^2 = 2 - 2A(\lambda, \nu), \tag{5}$$

where  $m$  dominates both measures. Let  $P_n$  select the first  $n$  coordinates and define the projected Hellinger energy

$$E_n(T) = -2 \log A((P_n)_\# \mu, (P_n)_\# \nu_T). \tag{6}$$

These are actual output marginals. They are not laws obtained from principal matrix corners.

**Theorem 1.1** (Complete criterion for arbitrary coordinate mixing). *Under (3), for every  $T \in \text{GL}(H)$ ,*

$$\boxed{\begin{aligned} \nu_T \sim \mu &\iff \exists Q \in \mathcal{P}_f : TQ^{-1} - I \in \mathcal{S}_2(H) \\ &\iff \sup_n E_n(T) < \infty. \end{aligned}} \tag{7}$$

*If these conditions fail, then  $\nu_T \perp \mu$  and  $E_n(T) \uparrow \infty$ . The exact bounded invertible linear stabilizer is  $\text{Stab}(\mu) = \mathcal{P}_f$ . On the equivalence branch, the projected likelihoods and reciprocal likelihoods are uniformly integrable in their respective reference measures.*

For any admissible  $Q$ , put  $K = TQ^{-1} - I$  and  $T_n = I + P_n K P_n$ . Then  $T_n$  is eventually invertible and

$$\frac{d\nu_{T_n}}{d\mu} \longrightarrow \frac{d\nu_{TQ^{-1}}}{d\mu} \quad \text{in } L^1(\mu), \quad \sqrt{\frac{d\nu_{T_n}}{d\mu}} \longrightarrow \sqrt{\frac{d\nu_{TQ^{-1}}}{d\mu}} \quad \text{in } L^2(\mu). \quad (8)$$

Since  $Q_{\#}\mu = \mu$ , the limiting law is also  $\nu_T$ .

The theorem contains the global energy formulation through  $\sup_n E_n(T)$ . The Fourier defect is not an additional scalar condition on  $T$ . It is the calibration mechanism. It turns small Hellinger displacement into squared Frobenius displacement. This distinction keeps the one-dimensional Gaussian characterization separate from the operator-valued equivalence criterion.

Two endpoint refinements separate the minimal assumptions cleanly.

**Corollary 1.2** (Moment-only stabilizer). *If the common coordinate law is centered, variance one, and non-Gaussian, then its bounded invertible linear stabilizer is  $\mathcal{P}_f$ . No density or Fisher-information assumption is needed.*

**Corollary 1.3** (Orthogonal endpoint). *If  $f > 0$  almost everywhere,  $\sqrt{f} \in W^{1,2}(\mathbb{R})$ , and  $f$  is centered, variance one, and non-Gaussian, then every orthogonal  $U$  obeys*

$$U_{\#}\mu \sim \mu \quad \Longleftrightarrow \quad U - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}_f.$$

Thus scale Fisher information is needed for general linear mixing but not for the orthogonal subproblem.

## 1.4 Organization and notation

Section 2 proves the global Fourier ODE theorem and explains the weighted-jet geometry. Section 3 constructs finite Fourier duals and establishes dimension-free local Hellinger coercivity; it also states the global coercivity theorem. Section 4 constructs the measurable action and proves the likelihood and zero-one machinery. Section 5 proves the Hilbert–Schmidt criterion, then uses a block-sum argument to complete the deferred globalization step. ?? gives exact operator-level applications and records the block and affine product criteria. Numerical illustrations are deferred to separate supplementary material.

| Symbol          | Meaning                                                                               |
|-----------------|---------------------------------------------------------------------------------------|
| $\phi$          | characteristic function $u \mapsto \mathbb{E}e^{iuX}$                                 |
| $R_\mu$         | rotational Fourier defect in (1)                                                      |
| $\Gamma_\mu$    | weighted Fourier jet $u \mapsto (u\phi(u), \phi'(u))$                                 |
| $A, h$          | Hellinger affinity and Hellinger distance                                             |
| $P_n, P_F$      | projections onto the first $n$ coordinates and a finite set $F$                       |
| $E_n(T)$        | projected Hellinger energy in (6)                                                     |
| $\mathcal{S}_2$ | Hilbert–Schmidt ideal; $\ \cdot\ _{\mathcal{S}_2} = \ \cdot\ _F$ in finite dimensions |
| $\Sigma_f$      | coordinate signs preserving the one-dimensional law                                   |
| $\mathcal{P}_f$ | coordinate permutations with signs in $\Sigma_f$                                      |
| $\Phi_T, \nu_T$ | measurable row-series action and its pushforward of $\mu$                             |

## 2 The Rotational Fourier Defect and Gaussian Characterization

The purpose of this section is to isolate the only genuinely one-dimensional obstruction in the paper. The detector is bounded. It uses no logarithm of the characteristic function. It remains valid in the presence of characteristic-function zeros. The conclusion belongs to the same broad lineage as the Kac–Bernstein theorem and the classical characterization theory of independent linear forms [7, 8]. Its hypothesis is different. An infinitesimal rotational Fourier response vanishes at every frequency pair. This leads directly to a global linear ODE.

### 2.1 The bivariate Fourier feature and cross-dependence isolation

This subsection requires only  $\mathbb{E}X = 0$  and  $\mathbb{E}|X| < \infty$ . Neither a density nor a second moment is needed. Write  $\phi(u) = \mathbb{E}e^{iuX}$ , and for independent copies  $X_1, X_2$  put

$$O_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}, \quad W_{\xi, \eta}(x, y) = (e^{i\xi x} - \phi(\xi))(e^{i\eta y} - \phi(\eta)).$$

This fixes the sign convention for rotation. Centering removes the two marginal responses. It leaves only the dependence created by rotation. In particular,

$$\mathbb{E}W_{\xi, \eta}(X_1, X_2) = 0, \quad \mathbb{E}|W_{\xi, \eta}(X_1, X_2)|^2 = (1 - |\phi(\xi)|^2)(1 - |\phi(\eta)|^2) \leq 1. \quad (9)$$

Pointwise,  $|W_{\xi, \eta}| \leq 4$ , so its variance under an arbitrary comparison law is at most 16. Both bounds are dimension-free and require no tail assumption.

### 2.2 A global linear ODE: zero defect characterizes Gaussians

**Lemma 2.1** (Logarithm-free characterization by rotational Fourier defect). *Let  $G_{\xi, \eta}(\theta) = \mathbb{E}W_{\xi, \eta}(O_\theta(X_1, X_2))$ . Then  $G_{\xi, \eta}(0) = 0$  and*

$$R_\mu(\xi, \eta) := G'_{\xi, \eta}(0) = \eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta). \quad (10)$$

Moreover,

$$R_\mu \equiv 0 \iff \text{Law}(X) \text{ is a centered Gaussian, possibly degenerate.} \quad (11)$$

*Proof.* The first moment implies  $\phi \in C^1(\mathbb{R})$ . The joint characteristic function after rotation is

$$\phi(\xi \cos \theta + \eta \sin \theta)\phi(-\xi \sin \theta + \eta \cos \theta).$$

The marginal characteristic functions are  $\phi(\xi \cos \theta)\phi(-\xi \sin \theta)$  and  $\phi(\eta \sin \theta)\phi(\eta \cos \theta)$ . Their derivatives at zero vanish because  $\phi'(0) = i\mathbb{E}X = 0$ . Differentiating the joint characteristic function therefore gives (10).

If  $R_\mu$  vanishes everywhere, continuity and  $\phi(0) = 1$  allow us to choose  $\eta_0 \neq 0$  with  $\phi(\eta_0) \neq 0$ . Fix this single frequency. The defect equation then becomes the global ODE

$$\phi'(\xi) = c\xi\phi(\xi), \quad c = \frac{\phi'(\eta_0)}{\eta_0\phi(\eta_0)}, \quad \phi(0) = 1.$$

No division by  $\phi(\xi)$  has occurred. Uniqueness for this linear ODE gives  $\phi(\xi) = \exp(c\xi^2/2)$  on all of  $\mathbb{R}$ , including any putative zero of  $\phi$ . The identity  $\phi(-\xi) = \overline{\phi(\xi)}$  forces  $c \in \mathbb{R}$ . Also  $|\phi(\xi)| \leq 1$  forces  $c \leq 0$ . Write  $c = -\sigma^2$ . The Gaussian conclusion follows. If  $\sigma = 0$ , we get the point mass at zero. Substitution proves the converse.  $\square$

### 2.3 Wedge-product structure and Fourier curvature

Define the weighted jet

$$\Gamma_\mu(u) = (u\phi(u), \phi'(u)) \in \mathbb{C}^2.$$

Then

$$\det \begin{pmatrix} s\phi(s) & \phi'(s) \\ t\phi(t) & \phi'(t) \end{pmatrix} = -R_\mu(s, t). \quad (12)$$

Thus the defect measures the change of projective direction of the weighted Fourier jet. In the Gaussian case  $\phi'(u) = -\sigma^2 u\phi(u)$ . So the jet lies in the fixed line spanned by  $(1, -\sigma^2)$ . Conversely, vanishing of every wedge product forces precisely the anchored ODE used above. This is the rigorous content of the “zero Fourier curvature” description.

For a non-Gaussian law, some two distinct frequencies have nonzero defect. By continuity, a compact neighborhood of this pair already detects the law. Thus a compact frequency set may be fixed after the coordinate law is chosen. This set is independent of the ambient dimension. The diagonal  $\xi = \eta$  always has zero defect and cannot serve this purpose. A global scalar energy is, for example,

$$\mathcal{R}_w(\mu) = \iint |R_\mu(\xi, \eta)|^2 w(\xi, \eta) d\xi d\eta, \quad w > 0 \text{ a.e.}, \quad \iint (1 + \xi^2 + \eta^2) w(\xi, \eta) d\xi d\eta < \infty. \quad (13)$$

Indeed,  $|R_\mu(\xi, \eta)| \leq \mathbb{E}|X|(|\xi| + |\eta|)$ . Hence this energy is finite and is zero exactly for centered Gaussian laws. Integrability of  $w$  alone need not make the energy finite.

The same feature gives a local Hellinger lower bound for one rotation. Choose a phase  $\alpha$  with  $\text{Re}(e^{-i\alpha} R_\mu(\xi, \eta)) > 0$  and apply

$$|\mathbb{E}_\lambda F - \mathbb{E}_\nu F| \leq h(\lambda, \nu) \{2\mathbb{E}_\lambda F^2 + 2\mathbb{E}_\nu F^2\}^{1/2}$$

to  $F = \text{Re}(e^{-i\alpha} W_{\xi, \eta})$ . It follows that  $h((O_\theta)_\# \text{Law}(X)^{\otimes 2}, \text{Law}(X)^{\otimes 2}) \geq c|\theta|$  for small  $|\theta|$ . A first moment suffices for this lower bound. The dimension-independent matrix estimates below require the additional second-moment and Sobolev hypotheses.

### 2.4 Dimension-free variance and affine normalization

Equation (9) is the variance estimate used for empirical or finite-frequency detection. Averaging  $N$  independent pairs gives standard error at most  $N^{-1/2}$  under the reference product. The rough bound  $16/N$  remains available under an arbitrary comparison distribution. In particular, the detector does not require a fourth moment, symmetry, or a density.

**Centering, scaling, and phase.** For an uncentered variable  $Y$  with mean  $m$ , use  $R_Y$  for the raw expression on the right of (10). Differentiating the centered feature now also differentiates its marginal terms. The actual dependence response is

$$D_Y(a, b) = R_Y(a, b) + im(a - b)\phi_Y(a)\phi_Y(b).$$

If  $X = (Y - m)/\sigma$ ,  $\sigma > 0$ , direct differentiation gives

$$R_X(\xi, \eta) = e^{-i(m/\sigma)(\xi+\eta)} D_Y(\xi/\sigma, \eta/\sigma). \quad (14)$$

Thus centering rotates the mean-corrected response by  $-(m/\sigma)(\xi + \eta)$ . Scaling also changes the frequencies. At fixed corresponding frequencies its modulus is unchanged. The raw expression  $R_Y$  alone does not transform by this phase factor. For a centered symmetric law the defect is real. At a particular frequency pair, an asymmetric law can still have a real or zero defect. So the phase is not, by itself, a complete measure of skewness.

For the concrete comparison  $Y \sim \text{Gamma}(2, 1)$ ,  $m = 2$ ,  $\sigma = \sqrt{2}$ , and  $(\xi, \eta) = (1, 2)$ , one obtains

$$D_Y(1/\sqrt{2}, \sqrt{2}) = \frac{4}{27}, \quad R_{(Y-2)/\sqrt{2}}(1, 2) = \frac{4}{27}e^{-3i\sqrt{2}}.$$

At this particular pair, the entire phase of the Gamma response is therefore the centering phase, namely  $-3\sqrt{2}$  modulo  $2\pi$ . The variance-one Laplace response is  $R_\mu(1, 2) = -4/27$ . Their moduli agree at these frequencies, while their principal phases are approximately  $116.915^\circ$  and  $180^\circ$ , respectively.

### 3 Finite-Dimensional Nonlinear Hellinger Coercivity

We now convert the one-dimensional Fourier obstruction into a matrix norm. The argument has two layers. The Fisher operator identifies the infinitesimal geometry. Finite Fourier sums provide bounded dual witnesses. Those norms do not grow with the ambient dimension. The resulting local estimate is then globalized modulo the discrete stabilizer in [Theorem 3.6](#).

#### 3.1 Fourier-to-Frobenius control and the matrix Fisher operator

Let  $f$  be a probability density and write  $g = \sqrt{f}$ . In this subsection we assume

$$\begin{aligned} f > 0 \quad \text{Lebesgue-almost everywhere,} \quad g \in W^{1,2}(\mathbb{R}), \quad xg'(x) \in L^2(\mathbb{R}), \\ \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad X \sim f(x) dx. \end{aligned} \tag{15}$$

The derivatives in these assumptions are weak derivatives. Define the location score  $s = -f'/f = -2g'/g$  almost everywhere, and put

$$\mathbb{E}s(X) = 0, \quad \mathbb{E}[Xs(X)] = 1, \quad J := \mathbb{E}s(X)^2 = 4\|g'\|_2^2, \tag{16}$$

and

$$d(x) := xs(x) - 1, \quad \kappa := \mathbb{E}d(X)^2 = 4\|xg'\|_2^2 - 1. \tag{17}$$

The two integration-by-parts identities in (16) follow from these assumptions. Indeed,  $f' = 2gg'$  belongs to  $L^1$ , and  $xf'$  belongs to  $L^1$  by Cauchy–Schwarz and the finite second moment. Applying integration by parts first to a compactly supported cutoff and then to  $x\chi(x/R)$  gives the identities as  $R \rightarrow \infty$ .

For  $X = (X_1, \dots, X_n) \sim \mu_n := f^{\otimes n}$  and  $A \in M_n(\mathbb{R})$ , the score of the pushforward by  $I + tA$  at  $t = 0$  is

$$S_A(X) = \sum_{i,j=1}^n A_{ij}(s(X_i)X_j - \delta_{ij}) = \sum_i A_{ii}d(X_i) + \sum_{i \neq j} A_{ij}s(X_i)X_j. \tag{18}$$

**Theorem 3.1** (The matrix Fisher operator). *Under (15), the quadratic form  $\mathcal{I}_f^{(n)}(A) := \mathbb{E}S_A^2$  has the decomposition*

$$\boxed{\mathcal{I}_f^{(n)} = \kappa P_{\text{diag}} + (J+1)P_{\text{sym,off}} + (J-1)P_{\text{skew}}.} \quad (19)$$

Here the projections are orthogonal for the Frobenius inner product. Equivalently, if  $D = \text{diag } A$ ,  $S_0 = (A + A^\top)/2 - D$ , and  $K = (A - A^\top)/2$ , then

$$\mathcal{I}_f^{(n)}(A) = \kappa \|D\|_F^2 + (J+1)\|S_0\|_F^2 + (J-1)\|K\|_F^2. \quad (20)$$

Their multiplicities are respectively

$$n, \quad n(n-1)/2, \quad n(n-1)/2. \quad (21)$$

Moreover,  $\kappa > 0$  and  $J \geq 1$ . Here  $J = 1$  holds precisely for the standard Gaussian density. Thus, for a non-Gaussian  $f$ ,

$$\lambda_f \|A\|_F^2 \leq \mathcal{I}_f^{(n)}(A) \leq C_f \|A\|_F^2, \quad \lambda_f := \min\{\kappa, J-1\} > 0, \quad C_f := \max\{\kappa, J+1\}. \quad (22)$$

*Proof.* The diagonal variables  $d(X_i)$  are independent and centered. Their covariances with every off-diagonal term  $s(X_j)X_k$ ,  $j \neq k$ , vanish. After any shared coordinate is integrated out, the remaining factor has expectation  $\mathbb{E}X$  or  $\mathbb{E}s(X)$ . Two off-diagonal terms can have a nonzero covariance only when their ordered index pairs coincide or are reversed. Those covariances are, respectively,  $J$  and  $(\mathbb{E}Xs(X))^2 = 1$ . Consequently

$$\mathbb{E}S_A^2 = \kappa \sum_i A_{ii}^2 + \sum_{i < j} \{J(A_{ij}^2 + A_{ji}^2) + 2A_{ij}A_{ji}\}. \quad (23)$$

Resolving each pair into its symmetric and skew-symmetric parts proves the decomposition. In particular, the calculation uses centering and independence and imposes no parity condition on  $f$ .

Cauchy–Schwarz applied to  $\mathbb{E}Xs(X) = 1$  gives  $J \geq 1$ . Equality implies  $s(X) = X$  almost surely, hence  $g' = -xg/2$  almost everywhere and  $f$  is the standard Gaussian. If  $\kappa = 0$ , then  $2xg' + g = 0$  almost everywhere on each of the two half-lines. The absolutely continuous solutions there are constant multiples of  $|x|^{-1/2}$ , which cannot give a positive probability density. This proves  $\kappa > 0$  and the asserted bounds.  $\square$

### 3.2 Hellinger paths, polar decomposition, and symmetric perturbations

For orientation, write the left polar decomposition of an invertible matrix as  $T = BO$ , where  $B = (TT^\top)^{1/2}$  is positive and  $O$  is orthogonal. The covariance defect is  $TT^\top - I = B^2 - I$ , so a finite second moment detects the symmetric factor. The skew tangent of the orthogonal factor has Fisher eigenvalue  $J-1$ , which is positive precisely off the Gaussian law. The full matrix Fisher decomposition above packages these two mechanisms without requiring a nonlinear separation of  $B$  and  $O$  at every step.

For  $A \in \text{GL}(n, \mathbb{R})$  define a unitary operator on  $L^2(\mathbb{R}^n, dx)$  by

$$(U_A v)(y) = |\det A|^{-1/2} v(A^{-1}y), \quad \Omega_n(x) = \prod_{i=1}^n g(x_i).$$



Thus  $U_A \Omega_n$  is the square root of the Lebesgue density of  $A_{\#} \mu_n$ , and  $h(A_{\#} \mu_n, B_{\#} \mu_n) = \|U_A \Omega_n - U_B \Omega_n\|_2$ . For a fixed real matrix  $A$ , the generator of  $U_{\exp(tA)}$  is

$$G_A v = -\frac{1}{2} \operatorname{tr}(A) v - (Ax) \cdot \nabla v, \quad G_A \Omega_n = \frac{1}{2} S_A \Omega_n. \quad (24)$$

These identities hold in  $L^2$ . To justify them under (15), observe that  $x_j \partial_i \Omega_n \in L^2$  for all  $i, j$ : the diagonal case uses  $xg' \in L^2$ , and the off-diagonal case uses  $g' \in L^2$  and  $\mathbb{E}X^2 = 1$ . Cutoff and mollification approximate  $\Omega_n$  in the corresponding weighted Sobolev graph norm, where the usual change-of-variables differentiation is valid.

**Lemma 3.2** (A dimension-free Hellinger path bound). *If  $A_t$ ,  $0 \leq t \leq 1$ , is a continuously differentiable path in  $\operatorname{GL}(n, \mathbb{R})$ , then*

$$h((A_1)_{\#} \mu_n, (A_0)_{\#} \mu_n) \leq \frac{\sqrt{C_f}}{2} \int_0^1 \|A_t^{-1} \dot{A}_t\|_F dt. \quad (25)$$

In particular, if  $\|B\|_{\operatorname{op}} \leq r < 1$ , then

$$h((I + B)_{\#} \mu_n, \mu_n) \leq \frac{\sqrt{C_f}}{2(1-r)} \|B\|_F. \quad (26)$$

All constants are independent of  $n$ .

*Proof.* The representation law  $U_A U_B = U_{AB}$  and (24) give

$$\frac{d}{dt} U_{A_t} \Omega_n = U_{A_t} G_{A_t^{-1} \dot{A}_t} \Omega_n, \quad \left\| \frac{d}{dt} U_{A_t} \Omega_n \right\|_2 = \frac{1}{2} \sqrt{\mathcal{I}_f^{(n)}(A_t^{-1} \dot{A}_t)}.$$

Integrate this identity and use the Fisher upper bound. For the last assertion take  $A_t = I + tB$  and use  $\|A_t^{-1}\|_{\operatorname{op}} \leq (1-r)^{-1}$ .  $\square$

### 3.3 Off-diagonal detection by finite Fourier projection

We now assume that  $f$  is non-Gaussian in addition to (15). We use finite real Fourier sums. These are finite real linear combinations of  $1$ ,  $\cos(\xi x)$  and  $\sin(\xi x)$  with real frequencies  $\xi$ . Their frequencies may depend on  $f$ . Once chosen, their finite set is independent of the matrix dimension.

**Lemma 3.3** (Bounded Fourier functions with exact dual moments). *There exist centered finite real Fourier sums  $p, q, r$  such that*

$$\begin{array}{c|cc} & s & x \\ \hline p & 1 & 0 \\ q & 0 & 1 \end{array} \quad \text{and} \quad \mathbb{E}[r(X)d(X)] = 1. \quad (27)$$

The entries in the table denote expectations of the products. For

$$H_A(X) := \sum_i A_{ii} r(X_i) + \sum_{i \neq j} A_{ij} p(X_i) q(X_j), \quad (28)$$

one has, for all real matrices  $A, B$  of the same size,

$$\mathbb{E}[H_A S_B] = \langle A, B \rangle_F, \quad \|H_A\|_{L^2(\mu_n)} \leq C_0 \|A\|_F, \quad (29)$$

where  $C_0$  is independent of  $n$ .

*Proof.* On the real vector space of centered finite real Fourier sums consider the two functionals

$$L_1(h) = \mathbb{E}h'(X) = \mathbb{E}[h(X)s(X)], \quad L_2(h) = \mathbb{E}[Xh(X)].$$

The integration-by-parts identity holds since  $h, h'$  are bounded. Both functionals are unchanged by subtracting a constant. If this moment map had rank at most one, all its two-by-two determinants would vanish, including after complexification. Writing  $\phi(\xi) = \mathbb{E}e^{i\xi X}$  and testing with the centered exponentials at frequencies  $\xi, \eta$  would then give

$$\eta\phi'(\xi)\phi(\eta) - \xi\phi(\xi)\phi'(\eta) = 0 \quad (\xi, \eta \in \mathbb{R}).$$

Here  $L_1(e^{i\xi x}) = i\xi\phi(\xi)$  and  $L_2(e^{i\xi x}) = -i\phi'(\xi)$ . The no-logarithm ODE argument in [Theorem 2.1](#) would force a centered Gaussian law, a contradiction. The map therefore has rank two. Inverting the moment matrix of two real Fourier witnesses supplies  $p, q$  with the displayed dual moments. This construction fixes finitely many frequencies depending only on  $f$ .

The diagonal function can be chosen using only the second moment. For  $\xi \neq 0$  let  $r_0(x) = \cos(\xi x) - \mathbb{E}\cos(\xi X)$ . Dominated convergence gives

$$\frac{\mathbb{E}[Xr_0'(X)]}{\xi^2} = -\mathbb{E}\left[X\frac{\sin(\xi X)}{\xi}\right] \rightarrow -\mathbb{E}X^2 = -1 \quad (\xi \rightarrow 0).$$

Choose a sufficiently small nonzero  $\xi$  and set  $r = r_0/\mathbb{E}[Xr_0'(X)]$ . Cutoff integration by parts gives  $\mathbb{E}[r(X)d(X)] = \mathbb{E}[Xr'(X)] = 1$ .

All diagonal/off-diagonal cross terms in  $\mathbb{E}H_A S_B$  vanish by centering. Coincident ordered off-diagonal pairs contribute  $\mathbb{E}[ps]\mathbb{E}[qX] = 1$ . Reversed pairs contribute  $\mathbb{E}[pX]\mathbb{E}[qs] = 0$ . This proves the pairing. For any centered  $a, b \in L^2(f)$ , independence gives

$$\left\| \sum_{i \neq j} A_{ij} a(X_i) b(X_j) \right\|_2 \leq \sqrt{2} \|a\|_2 \|b\|_2 \|A\|_F. \quad (30)$$

Together with the independent diagonal terms, this proves the last assertion.  $\square$

The next lemma records the uniform estimate needed to use these detectors along a matrix path. Its proof gives the index contractions explicitly. In particular, it uses no fourth moment of  $X$  or higher moment of the score.

**Lemma 3.4** (Generator bound for bounded coordinate detectors). *Let  $p, q, r$  be any fixed centered bounded continuously differentiable functions with bounded derivatives. Define  $H_A$  by (28). There is a constant  $C_1$ , depending on these functions and on  $J, \kappa$ , such that*

$$\|G_B(H_A\Omega_n)\|_{L^2(dx)} \leq C_1 \|A\|_F \|B\|_F \quad (A, B \in M_n(\mathbb{R}), n \geq 1). \quad (31)$$

*Proof.* For fixed  $n$ , both  $H_A$  and its first derivatives are bounded. Consequently

$$x_j \partial_i (H_A \Omega_n) = H_A x_j \partial_i \Omega_n + x_j (\partial_i H_A) \Omega_n \in L^2(dx)$$

by the weighted Sobolev regularity of  $\Omega_n$  and  $\mathbb{E}X^2 = 1$ . The cutoff and mollification argument used for (24) therefore puts  $H_A \Omega_n$  in the domain of  $G_B$ . The product rule gives

$$G_B(H_A \Omega_n) = \left\{ \frac{1}{2} H_A S_B - (Bx) \cdot \nabla H_A \right\} \Omega_n. \quad (32)$$

Both terms belong to  $L^2$  in each finite dimension. We prove a dimension-independent bound for each term.

We will repeatedly use the following elementary consequence of independence. If  $h_1, \dots, h_m$  are centered  $L^2(f)$  functions, and a coefficient tensor  $C$  is indexed by ordered tuples of *distinct* coordinates, then

$$\left\| \sum_{i_1, \dots, i_m}^{\neq} C_{i_1 \dots i_m} \prod_{a=1}^m h_a(X_{i_a}) \right\|_2 \leq \sqrt{m!} \|C\|_{\ell^2} \prod_{a=1}^m \|h_a\|_2. \quad (33)$$

In the square of this norm, a mixed term vanishes unless the two tuples have the same index set. There are at most  $m!$  matching permutations. Cauchy–Schwarz for the local moments gives a bound on each index pattern. A second Cauchy–Schwarz step for the coefficient arrays proves the full bound. We use it only for  $m \leq 4$ .

First consider  $H_A S_B$ . For a vector  $a$  and a zero-diagonal matrix  $M$ , abbreviate

$$D_a(u) = \sum_i a_i u(X_i), \quad Q_M(u, v) = \sum_{i \neq j} M_{ij} u(X_i) v(X_j).$$

The product  $H_A S_B$  is the sum of the four products obtained by multiplying  $D_{\mathbf{a}}(r) + Q_{A_0}(p, q)$  and  $D_{\mathbf{b}}(d) + Q_{B_0}(s, x)$ , where  $\mathbf{a} = (A_{ii})_i$ ,  $\mathbf{b} = (B_{ii})_i$ ,  $A_0 = A - \text{diag}(\mathbf{a})$ , and  $B_0 = B - \text{diag}(\mathbf{b})$ . In every common coordinate, a local product is a bounded function times one of  $d, s, x$  and hence is in  $L^2(f)$ . Write each such local product as its mean plus its centered part.

For a product  $D_a(u) D_b(v)$ , distinct indices give a coefficient tensor bounded by  $\|a\|_2 \|b\|_2$ ; equal indices give the Hadamard vector  $(a_i b_i)_i$  and the scalar  $\sum_i a_i b_i$ . Their norms have the same bound. For  $D_a(u) Q_M(v, w)$ , there are three index patterns: three distinct indices, equality with the first index of  $Q_M$ , and equality with its second index. The first gives the tensor  $a \otimes M$ . In either equality case, centering the local product gives a matrix obtained by row or column multiplication of  $M$  by  $a$ , and its mean gives  $M^T a$  or  $M a$ . Each norm is at most  $\|a\|_2 \|M\|_F$ . The product  $Q_M(u, v) D_a(w)$  has exactly the same three patterns and bounds.

For clarity, all patterns in a quadratic–quadratic product can also be enumerated. Consider

$$Q_M(u, v) Q_N(w, z) = \sum_{i \neq j, k \neq l} M_{ij} N_{kl} u_i v_j w_k z_l,$$

where  $u, v$  are bounded and  $w, z$  are in  $L^2$ . Four distinct indices give the restricted tensor  $M \otimes N$ , of norm at most  $\|M\|_F \|N\|_F$ . There are four patterns with exactly one shared index:  $i = k$ ,  $i = l$ ,  $j = k$ , and  $j = l$ . For example, the pattern  $i = k$  is

$$\sum_{i, j, l}^{\neq} M_{ij} N_{il} (uw)_i v_j z_l.$$

Its centered local product has coefficient norm bounded by

$$\left\{ \sum_i \left( \sum_j M_{ij}^2 \right) \left( \sum_l N_{il}^2 \right) \right\}^{1/2} \leq \|M\|_F \|N\|_F.$$

Its local mean has coefficient matrix  $(M^T N)_{jl}$  off the diagonal; excluding  $i = j, l$  does not change the sum because  $M, N$  have zero diagonal. The other three shared-index patterns give the same

estimate with a transpose or with the two coordinates exchanged, and their mean contractions are matrix products of  $M, N$  or their transposes.

Finally, there are two patterns with both indices shared:  $(k, l) = (i, j)$  and  $(k, l) = (j, i)$ . Their coefficients are  $M_{ij}N_{ij}$  and  $M_{ij}N_{ji}$ . After centering the two local products, the resulting coefficients are a Hadamard matrix, its row sums, its column sums, and its total sum. For instance

$$\sum_i \left| \sum_j M_{ij}N_{ij} \right|^2 \leq \sum_i \left( \sum_j M_{ij}^2 \right) \left( \sum_j N_{ij}^2 \right) \leq \|M\|_F^2 \|N\|_F^2,$$

and the other three estimates follow from the same Cauchy–Schwarz inequality. These cases exhaust the possible index coincidences. Applying (33) to the centered pieces therefore proves

$$\|H_A S_B\|_2 \leq C \|A\|_F \|B\|_F. \quad (34)$$

It remains to bound  $(Bx) \cdot \nabla H_A$ . We give the contractions separately, since derivatives of a centered function need not be centered. For the diagonal part the sum is

$$\sum_{i,k} A_{ii} B_{ik} r'(X_i) X_k.$$

When  $i \neq k$ , split  $r'$  into its mean and centered part. The latter has coefficient matrix  $\text{diag}(\mathbf{a})B$ , with its diagonal deleted. The mean has coefficient vector  $B^\top \mathbf{a} - \mathbf{a} \odot \mathbf{b}$ . Their norms are bounded by, respectively, one and two times  $\|A\|_F \|B\|_F$ . When  $i = k$ , center the local function  $xr'(x)$ , which is in  $L^2$  since  $r'$  is bounded. The coefficient vector and the mean are bounded by the same product of matrix norms.

Let  $M = A_0$ . The derivative of the off-diagonal part is the sum of

$$F(M, B; p', q) := \sum_{i \neq j, k} M_{ij} B_{ik} p'(X_i) q(X_j) X_k$$

and  $F(M^\top, B; q', p)$ . In  $F$ , first take  $k$  distinct from  $i, j$  and split  $p'$  into its mean and centered part. The centered part has three-index coefficient norm bounded by

$$\left\{ \sum_i \left( \sum_j M_{ij}^2 \right) \left( \sum_k B_{ik}^2 \right) \right\}^{1/2} \leq \|M\|_F \|B\|_F.$$

The mean part has, off the diagonal, the matrix  $M^\top B - M^\top \text{diag}(\mathbf{b})$ ; the second term removes  $i = k$ , and  $i = j$  already contributes zero. Its Frobenius norm is at most  $2\|M\|_F \|B\|_F$ . When  $k = i$ , center  $xp'(x)$ ; the resulting matrix has entries  $B_{ii}M_{ij}$  and its mean has vector  $M^\top \mathbf{b}$ . When  $k = j$ , split both  $p'$  and  $xq(x)$  into means and centered parts; the coefficients are the Hadamard matrix  $M \odot B$ , its row and column sums, and its total sum. The preceding Hadamard estimates bound all of them. All the local functions used here belong to  $L^2$ , since  $p', q$  are bounded and  $\mathbb{E}X^2 = 1$ . The same proof applies to  $F(M^\top, B; q', p)$ . Using (33) again proves

$$\|(Bx) \cdot \nabla H_A\|_2 \leq C \|A\|_F \|B\|_F.$$

Together with (34) and (32), this proves the lemma.  $\square$

**Proposition 3.5** (Uniform local Hellinger coercivity). *For every non-Gaussian density satisfying (15), there are constants  $c, C, \delta > 0$ , independent of  $n$ , such that*

$$c\|B\|_F \leq h((I + B)_\# \mu_n, \mu_n) \leq C\|B\|_F \quad \text{whenever } \|B\|_F \leq \delta. \quad (35)$$

The lower bound can be obtained using the fixed finite set of coordinate frequencies from [Theorem 3.3](#).

*Proof.* Fix the dual functions from [Theorem 3.3](#). First use exponential coordinates. Unitarity and the one-parameter group derivative give

$$\begin{aligned} \langle H_A \Omega_n, (U_{\exp A} - I) \Omega_n \rangle &= \int_0^1 \langle U_{\exp(-tA)}(H_A \Omega_n), G_A \Omega_n \rangle dt \\ &= \frac{1}{2} \|A\|_F^2 + \mathcal{R}_A. \end{aligned}$$

Here the exact leading term is  $\frac{1}{2} \mathbb{E} H_A S_A$ . For a vector in the domain of  $G_A$ , the fundamental theorem of calculus gives  $\|(U_{\exp(-tA)} - I)v\|_2 \leq t\|G_A v\|_2$ . Thus [Theorem 3.4](#) and the Fisher upper bound imply

$$|\mathcal{R}_A| \leq \frac{C_1 \sqrt{C_f}}{4} \|A\|_F^3.$$

Cauchy–Schwarz and (29) now give, for all sufficiently small  $\|A\|_F$ ,

$$h((\exp A)_\# \mu_n, \mu_n) \geq \frac{1}{4C_0} \|A\|_F.$$

For  $\|B\|_{\text{op}} < 1/2$ , the real power-series logarithm  $A = \log(I + B)$  exists and satisfies

$$\|A - B\|_F \leq \sum_{k \geq 2} \frac{\|B\|_{\text{op}}^{k-1}}{k} \|B\|_F \leq C\|B\|_{\text{op}} \|B\|_F.$$

Its norm is therefore uniformly comparable to  $\|B\|_F$  in a fixed small chart. This proves the lower bound for  $I + B$ ; the upper bound follows from [Theorem 3.2](#).  $\square$

The preceding pairing is a half-density pairing: it is  $\int H_A \sqrt{d\mu_n} d((\exp A)_\# \mu_n) - \int H_A d\mu_n$ . It provides a uniform local lower bound without asserting a dimension-uniform variance bound for  $H_A$  under every arbitrarily mixed output law. Such a bound, or an independent tail argument, is a separate requirement when passing to necessity for arbitrary output marginals.

### 3.4 Uniform coercivity modulo the discrete stabilizer

Let  $\mathcal{P}_f(n)$  denote the allowed signed-permutation matrices in dimension  $n$ , and set

$$d_{n,f}(T) = \min_{Q \in \mathcal{P}_f(n)} \|T - Q\|_F. \quad (36)$$

**Theorem 3.6** (Dimension-free global Hellinger coercivity). *Assume (3). For every  $0 < a \leq b < \infty$  there are constants  $c = c(f, a, b) > 0$  and  $C = C(f, a, b) < \infty$  such that, for every  $n$  and every  $T \in \text{GL}(n, \mathbb{R})$  whose singular values lie in  $[a, b]$ ,*

$$c \min\{d_{n,f}(T)^2, 1\} \leq h(\mu_n, T_\# \mu_n)^2 \leq C \min\{d_{n,f}(T)^2, 1\}. \quad (37)$$

The local regime of (37) is exactly [Theorem 3.5](#), after conjugating by a member of  $\mathcal{P}_f(n)$ . The uniform separation away from that local chart is a genuinely global statement. Its proof, given in [Section 5.5](#), uses the infinite-dimensional equivalence theorem as a compactness device: a hypothetical sequence of distant near-invariances is assembled into one block-diagonal equivalence, which the Hilbert–Schmidt classification rules out. The global estimate is therefore a consequence of the main theorem and is not used in its proof.

## 4 Infinite-Dimensional Measurable Realization and Dichotomy

The sample space is  $\mathbb{R}^{\mathbb{N}}$ , not  $\ell^2$ , so a bounded operator on  $\ell^2$  does not act pointwise by a naively convergent matrix product. This section constructs the required measurable versions. Then it identifies the global measure class through likelihood martingales and a dense translation group.

### 4.1 Measurable linear actions and countable-orbit realizations

Throughout this subsection, only  $\mathbb{E}_f X = 0$  and  $\mathbb{E}_f X^2 = 1$  are needed. Put  $\mu = f^{\otimes \mathbb{N}}$  on  $X = \mathbb{R}^{\mathbb{N}}$ , and  $\mu_n = f^{\otimes n}$ . The sample sequence is not an  $\ell^2$ -valued random variable. So we first specify how an operator on  $\ell^2$  acts on its law.

**Lemma 4.1** (Measurable realization on each countable operator orbit). *For every  $T \in \mathcal{B}(\ell^2)$ , there is a Borel map  $\Phi_T : X \rightarrow X$  whose coordinates are the  $L^2(\mu)$  row sums*

$$(\Phi_T x)_i = \sum_{j \geq 1} T_{ij} x_j.$$

*These maps can be chosen so that the following statements hold. If a probability measure  $\lambda$  satisfies*

$$\int \left| \sum_j a_j x_j \right|^2 d\lambda(x) \leq C_\lambda \|a\|_2^2 \quad \text{for every finitely supported } a, \quad (38)$$

*then  $(\Phi_T)_\# \lambda$  also satisfies this condition. Its constant is at most  $C_\lambda \|T\|^2$ . Also*

$$\Phi_S \circ \Phi_T = \Phi_{ST} \quad \lambda\text{-almost surely} \quad (S, T \in \mathcal{B}(\ell^2)). \quad (39)$$

*For each fixed countable subgroup  $\Gamma \leq \text{GL}(\ell^2)$ , there is a  $\Gamma$ -invariant Borel set  $D_\Gamma \subset X$  of full measure for every  $\lambda$  satisfying (38). On this set, these restrictions form an action by bimeasurable automorphisms. In particular, this applies to  $\mu$  and all its  $\Gamma$ -orbit laws, including singular orbit laws.*

*Proof.* For  $a \in \ell^2$ , the finite sums  $\sum_{j \leq n} a_j x_j$  have a unique  $L^2(\lambda)$  limit  $L_a$ , by (38). Choose

$$n_k(a) = \min \left\{ n \geq k : \sum_{j > n} |a_j|^2 \leq 2^{-4k} \right\}.$$

Define  $\widehat{L}_a(x)$  to be the limit of  $\sum_{j \leq n_k(a)} a_j x_j$  when it exists, and zero otherwise. The estimate

$$\left\| \sum_{j \leq n_k(a)} a_j x_j - L_a \right\|_{L^2(\lambda)}^2 \leq C_\lambda 2^{-4k}$$

and Borel–Cantelli show that  $\widehat{L}_a = L_a$ ,  $\lambda$ -almost surely. This one definition therefore works for every law satisfying (38); it does not depend on knowing whether that law is equivalent to  $\mu$ . Set  $\Phi_T(x)_i = \widehat{L}_{T^*e_i}(x)$ . The construction is Borel, and the same is true jointly in the matrix coefficients of  $T$  and in  $x$ , since each tail norm and the integer choice  $n_k$  is Borel.

For finitely supported  $b$ , the finite sum of output coordinates equals  $L_{T^*b}$  almost surely. This proves the asserted second-moment bound for the pushforward. For the  $i$ -th row of  $S$ , finite output sums satisfy

$$\sum_{j \leq n} S_{ij} \widehat{L}_{T^*e_j} = L_{T^*P_n S^*e_i} \longrightarrow L_{T^*S^*e_i} \quad \text{in } L^2(\lambda).$$

The selected subsequence on the left also converges almost surely to  $\Phi_S(\Phi_T(x))_i$ , because the output law has the required second-moment bound. Uniqueness of the limit proves (39).

For a countable  $\Gamma$ , let  $B$  be the Borel set on which all its row limits, identity relations, and pairwise composition relations hold. The preceding argument gives  $\lambda(B) = 1$  for every law under consideration. Put

$$D_\Gamma = \bigcap_{T \in \Gamma} \Phi_T^{-1}(B).$$

Every orbit law satisfies the same second-moment condition, so  $\lambda(D_\Gamma) = 1$ . The composition identities imply that  $D_\Gamma$  is invariant and that  $\Phi_{T^{-1}}$  is the inverse of  $\Phi_T$  there. This proves the final assertion.  $\square$

Write  $\nu_T = (\Phi_T)_\# \mu$ . The preceding lemma makes assertions about a common carrier only for a fixed countable subgroup. No single pointwise action of all of  $\text{GL}(\ell^2)$  on a common conull set is required. The subsequences in its proof are also essential. A covariance bound gives convergence of arbitrary row sums in  $L^2$ . That is not enough for almost-sure convergence of all ordinary partial sums under a correlated orbit law. On the original independent product law, the versions agree with the usual almost-sure row sums.

## 4.2 Projected likelihood martingales and Hellinger energy

**Lemma 4.2** (Projected likelihoods and two-sided uniform integrability). *For  $T \in \text{GL}(\ell^2)$ , every finite output law  $\nu_{T,N} = (P_N)_\# \nu_T$  is equivalent to  $\mu_N$ . Writing  $\mathcal{F}_N = \sigma(x_1, \dots, x_N)$ , define*

$$R_N(x) = \frac{d\nu_{T,N}}{d\mu_N}(x_{1:N}), \quad \mathbf{A}_N(T) = \mathbb{E}_\mu \sqrt{R_N}, \quad E_N(T) = -2 \log \mathbf{A}_N(T). \quad (40)$$

*Then  $(R_N)$  is a nonnegative mean-one  $\mu$ -martingale. We have*

$$\mathbf{A}_N(T) \downarrow \mathbf{A}(\mu, \nu_T), \quad \nu_T \sim \mu \iff \sup_N E_N(T) < \infty. \quad (41)$$

*Otherwise  $E_N(T) \uparrow \infty$ . Then  $\nu_T \perp \mu$ . If  $\nu_T \sim \mu$  with density  $Z_T$ , then*

$$R_N = \mathbb{E}_\mu[Z_T \mid \mathcal{F}_N] \longrightarrow Z_T \quad \text{in } L^1(\mu), \quad (42)$$

$$R_N^{-1} = \mathbb{E}_{\nu_T}[Z_T^{-1} \mid \mathcal{F}_N] \longrightarrow Z_T^{-1} \quad \text{in } L^1(\nu_T). \quad (43)$$

*In particular, both likelihood martingales are uniformly integrable in their respective reference measures.*

*Proof.* The first  $N$  rows of  $T$  are linearly independent, since  $T^*$  is injective. Some  $N$  input columns therefore form an invertible  $N \times N$  minor  $A$ . Split the input into these coordinates and their complement. The first  $N$  outputs have the form  $AU + W$ , where  $U \sim \mu_N$  and  $W$  is independent of  $U$ . Conditioning on  $W$  writes their density as a mixture of translates of the strictly positive density of  $AU$ . This mixture is positive and finite almost everywhere, proving the finite-dimensional equivalence.

Consistency of projected laws proves the martingale assertion. The martingale limit is  $R_\infty = d(\nu_T)_a/d\mu$ , the density of the absolutely continuous part of  $\nu_T$ . To see the identification explicitly, Fatou's lemma on cylinder sets gives  $R_\infty \mu \leq \nu_T$ , first on the cylinder algebra and then on the generated sigma-field. Conversely, if  $(\nu_T)_a = Z\mu$ , then  $\mathbb{E}_\mu[Z \mid \mathcal{F}_N] \leq R_N$ , so martingale convergence gives  $Z \leq R_\infty$ . The two inequalities identify the limit. Since  $\mathbb{E}_\mu(\sqrt{R_N})^2 = 1$ , the square roots are uniformly integrable in  $L^1(\mu)$ . They thus converge in  $L^1$  to  $\sqrt{R_\infty}$ . Conditional Jensen's inequality gives the monotonicity of their expectations, and consequently

$$\lim_N A_N(T) = \int \sqrt{R_\infty} d\mu = A(\mu, \nu_T).$$

The dichotomy in [Theorem 4.3](#) now proves (41) and the singular alternative.

Under equivalence, the forward conditional-expectation identity is the definition of a projected density. The reverse identity follows because for  $B \in \mathcal{F}_N$ ,

$$\int_B R_N^{-1} d\nu_T = \mu(B) = \int_B Z_T^{-1} d\nu_T.$$

Conditional-expectation convergence gives both  $L^1$  limits and hence both uniform-integrability assertions.  $\square$

For an arbitrary finite  $F \subseteq \mathbb{N}$ , let  $P_F$  be the corresponding coordinate projection and define

$$E_F(\mu, \nu) = -2 \log A((P_F)_\# \mu, (P_F)_\# \nu).$$

Data processing makes  $E_F$  monotone under inclusion. The initial segments  $\{1, \dots, n\}$  are cofinal among finite subsets.

$$\sup_F E_F = \sup_n E_n.$$

Thus the energy criterion does not depend on a privileged enumeration. Initial segments are still convenient for the martingale notation.

### 4.3 Rational translation invariance and the zero-one dichotomy

**Lemma 4.3** (Dichotomy for every bounded invertible orbit). *For every  $T \in \text{GL}(\ell^2)$ ,*

$$\nu_T \sim \mu \quad \text{or} \quad \nu_T \perp \mu. \tag{44}$$

*This conclusion uses only positivity of  $f$ , its centered finite second moment, and  $\sqrt{f} \in W^{1,2}(\mathbb{R})$ . In particular, it holds under (3).*

*Proof.* Put  $g = \sqrt{f}$ . The translation estimate

$$1 - A(f, f(\cdot - a)) = \frac{1}{2} \|g(\cdot - a) - g\|_2^2 \leq \frac{1}{2} \|g'\|_2^2 a^2$$



and Kakutani's theorem imply  $(\tau_h)_\# \mu \sim \mu$  for every  $h \in \ell^2$ . The row realizations satisfy

$$\Phi_T(x + T^{-1}h) = \Phi_T(x) + h \quad \mu\text{-almost surely};$$

this follows directly by adding the absolutely convergent scalar row pairing with the  $\ell^2$  vector  $T^{-1}h$ . Consequently,  $\nu_T$  is also quasi-invariant under every  $\ell^2$  translation.

Let  $H$  be the countable additive group generated by  $\{qe_j, qTe_j : q \in \mathbb{Q}, j \geq 1\}$ . Both  $\mu$  and  $\nu_T$  are quasi-invariant and ergodic under translation by  $H$ . Indeed, an event invariant modulo  $\mu$  under all rational finite-coordinate translations has, by positivity of the finite-dimensional densities, almost-everywhere constant sections in each finite coordinate block. It is therefore a tail event modulo  $\mu$ , and has probability zero or one. Here one uses the elementary fact that a measurable function on  $\mathbb{R}^n$  invariant almost everywhere under rational translations is constant almost everywhere, obtained, for example, by convolution with smooth approximate identities. For  $\nu_T$ , pull an invariant event back by  $\Phi_T$ ; invariance under  $qTe_j$  becomes invariance under  $qe_j$ , giving the same zero-one conclusion.

Write  $\nu_T = \nu_a + \nu_s$  for its Lebesgue decomposition relative to  $\mu$ . Choose a Borel  $\mu$ -null set  $N_0$  supporting  $\nu_s$ , and let  $N = \bigcup_{h \in H} (N_0 + h)$ . Quasi-invariance of  $\mu$  makes  $N$  a  $\mu$ -null set; it is exactly  $H$ -invariant and still supports  $\nu_s$ . Thus  $\nu_T(N) = \nu_s(X)$ , and ergodicity of  $\nu_T$  makes this number zero or one. The latter case gives singularity. In the former case, write  $\nu_T = Z\mu$ . Quasi-invariance of both measures implies that  $\{Z > 0\}$  is invariant modulo  $\mu$  under  $H$ . Its  $\mu$ -measure is positive because  $\int Z d\mu = 1$ , and ergodicity of  $\mu$  makes that measure one. This gives equivalence.  $\square$

## 5 Proof of the Measure Equivalence Theorem

We now assemble the finite-dimensional coercivity and the measurable zero-one machinery. Only the local coercivity theorem from [Section 3](#) is used in the necessity argument. The global finite-dimensional estimate is proved afterward.

### 5.1 Sufficiency of Hilbert–Schmidt perturbations

**Proposition 5.1** (Hilbert–Schmidt sufficiency and RN closure). *Assume (15), let  $\mu = f^{\otimes \mathbb{N}}$ , and let  $T = I + K \in \text{GL}(\ell^2)$  with  $K \in \mathcal{S}_2$ . Then  $\nu_T \sim \mu$ . For the eventually invertible operators  $T_n = I + P_n K P_n$ , their likelihoods  $Z_n = d\nu_{T_n}/d\mu$  converge in  $L^1(\mu)$  to the strictly positive likelihood  $Z_T = d\nu_T/d\mu$ , and  $\sqrt{Z_n} \rightarrow \sqrt{Z_T}$  in  $L^2(\mu)$ . The inverse approximants satisfy the same conclusion:*

$$\frac{d\nu_{T_n^{-1}}}{d\mu} \longrightarrow \frac{d\nu_{T^{-1}}}{d\mu} \quad \text{in } L^1(\mu).$$

*On the common measurable carrier of the relevant countable operator group, the likelihoods obey*

$$Z_{ST}(y) = Z_S(y)Z_T(\Phi_{S^{-1}}y), \tag{45}$$

$$Z_T(y)Z_{T^{-1}}(\Phi_{T^{-1}}y) = 1. \tag{46}$$

*If  $Q$  is an exact linear stabilizer of  $\mu$ , the same equivalence holds whenever  $TQ^{-1} - I \in \mathcal{S}_2$  and  $T$  is invertible.*

*Proof.* Since  $P_n K P_n \rightarrow K$  in Hilbert–Schmidt norm,  $T_n \rightarrow T$  in operator norm. The approximants are therefore eventually invertible. Their inverse norms are uniformly bounded. If  $A_n$  is the first  $n$ -coordinate block of  $T_n$ , then

$$Z_n(x) = |\det A_n|^{-1} \prod_{i=1}^n \frac{f((A_n^{-1} x_{1:n})_i)}{f(x_i)}. \quad (47)$$

These densities are positive almost everywhere. For all sufficiently large  $m, n$ , put  $B_{nm} = T_n^{-1} T_m - I$ . It acts in a finite coordinate block. Its operator norm is at most  $1/2$ . Hellinger invariance under the finite invertible map  $T_n^{-1}$  and (26) give

$$\|\sqrt{Z_m} - \sqrt{Z_n}\|_2 \leq \sqrt{C_f} \|T_n^{-1}\|_{\text{op}} \|T_m - T_n\|_{\mathcal{S}_2}. \quad (48)$$

Thus the square roots are Cauchy in  $L^2(\mu)$ . They have a nonnegative limit  $r$  of norm one. Cauchy–Schwarz also gives

$$\|Z_m - Z_n\|_1 \leq 2 \|\sqrt{Z_m} - \sqrt{Z_n}\|_2, \quad Z_n \rightarrow r^2 \quad \text{in } L^1(\mu). \quad (49)$$

For each fixed finite set of output coordinates, the row-series realizations  $\Phi_{T_n} X$  converge to  $\Phi_T X$  in  $L^2$ . Bounded continuous cylinder tests identify  $r^2 \mu$  with  $\nu_T$ .

The inverse approximants converge in Hilbert–Schmidt norm because

$$\|T_n^{-1} - T^{-1}\|_{\mathcal{S}_2} \leq \|T_n^{-1}\|_{\text{op}} \|T_n - T\|_{\mathcal{S}_2} \|T^{-1}\|_{\text{op}}.$$

Apply the same argument to these finite-block operators. This gives  $\nu_{T^{-1}} \ll \mu$ . The measurable row-series construction in Theorem 4.1 requires only the centered finite second moment. On its countable-group carrier it gives

$$\mu = (\Phi_T)_\# \nu_{T^{-1}} \ll (\Phi_T)_\# \mu = \nu_T.$$

This proves equivalence and strict positivity of  $r^2$ . Change of variables on the same carrier proves the cocycle and inverse identities. Finally,  $\nu_Q = \mu$  and  $T = (TQ^{-1})Q$  give the stabilizer assertion.  $\square$

### 5.1.1 The orthogonal case requires only location Fisher information

For orthogonal transformations the assumptions can be reduced to

$$f > 0 \text{ almost everywhere,} \quad g = \sqrt{f} \in W^{1,2}(\mathbb{R}), \quad \mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1. \quad (50)$$

In particular,  $xg' \in L^2$  is not required. The cutoff identities  $\mathbb{E}s = 0$  and  $\mathbb{E}Xs = 1$  still follow from  $J < \infty$  and the second moment. For a skew-symmetric matrix  $A$ , the score has only off-diagonal terms and

$$\mathbb{E}S_A^2 = (J - 1) \|A\|_F^2. \quad (51)$$

The generator formula remains valid. Only  $x_j \partial_i \Omega_n$  with  $i \neq j$  occur. One can justify it by radial cutoff and radial mollification. Those operators commute with the orthogonal one-parameter group.

**Proposition 5.2** (Orthogonal local bounds and HS sufficiency). *Under (50), every orthogonal  $U$  on  $\ell^2$  with  $U - I \in \mathcal{S}_2$  satisfies  $\nu_U \sim \mu$ . Its likelihoods have the same  $L^1$  and square-root  $L^2$  closure along finite-coordinate orthogonal approximants, as do the inverse likelihoods. If  $f$  is non-Gaussian, there also exist dimension-independent  $c, C, \delta > 0$  such that*

$$c\|Q - I\|_F \leq h(Q_{\#}\mu_n, \mu_n) \leq C\|Q - I\|_F \quad (Q \in O(n), \|Q - I\|_F \leq \delta). \quad (52)$$

*Proof.* For a skew-symmetric  $A$ , integration along  $\exp(tA)$  gives

$$h((\exp A)_{\#}\mu_n, \mu_n) \leq \frac{1}{2}\sqrt{J-1}\|A\|_F. \quad (53)$$

If  $f$  is non-Gaussian, the construction of the two dual functions  $p, q$  in Theorem 3.3 uses only  $s, X \in L^2$  and  $J - 1 > 0$ . On skew-symmetric matrices use  $H_A = \sum_{i \neq j} A_{ij}p(X_i)q(X_j)$ , with no diagonal function. The score  $S_B$  likewise has no diagonal term. Consequently the proof of Theorem 3.4 contains only bounded functions times  $s$  or  $X$ , and proves the same estimate under (50). The half-density pairing argument gives the local lower bound in exponential coordinates. For an orthogonal  $Q$  sufficiently close to  $I$ , its power-series logarithm is real and skew-symmetric, with  $\|\log Q\|_F$  comparable to  $\|Q - I\|_F$ . This proves (52).

For completeness, finite-coordinate orthogonal approximants can be constructed without applying a nonorthogonal path estimate. Set  $T_n = I + P_n(U - I)P_n$  and, for large  $n$ , take its polar factor

$$Q_n = T_n(T_n^{\top}T_n)^{-1/2}.$$

It is orthogonal and acts identically outside the first  $n$  coordinates. Since  $T_n \rightarrow U$  in Hilbert–Schmidt norm,

$$\|T_n^{\top}T_n - I\|_{\mathcal{S}_2} \leq (\|T_n\|_{\text{op}} + 1)\|T_n - U\|_{\mathcal{S}_2} \rightarrow 0.$$

The smallest singular values of  $T_n$  are bounded below by some  $m > 0$ . Functional calculus for  $|T_n|$  then gives

$$\|Q_n - T_n\|_{\mathcal{S}_2} = \|I - |T_n|\|_{\mathcal{S}_2} \leq \frac{\|T_n^{\top}T_n - I\|_{\mathcal{S}_2}}{1 + m}.$$

Thus  $Q_n \rightarrow U$  in Hilbert–Schmidt norm. For sufficiently large  $n, m$ ,  $Q_n^{\top}Q_m$  is close to the identity and has a small skew-symmetric logarithm. Applying (53) to this logarithm proves that the likelihood square roots are Cauchy. Their  $L^1$  limits are identified by the same cylinder tests as in Theorem 5.1. Apply the argument also to  $Q_n^{-1} \rightarrow U^{-1}$ ; the measurable inverse action then proves equivalence and the asserted closure.  $\square$

## 5.2 Necessity: tail rigidity and global Hilbert–Schmidt collapse

The next lemma transports finite-dimensional local coercivity to genuine Hilbert–Schmidt perturbations, including finite-rank maps with infinitely supported rows. It is the quantitative input to the tail-commutator argument.

**Lemma 5.3** (The local bound for Hilbert–Schmidt operators). *Let  $f$  be non-Gaussian and satisfy (15). There are constants  $c, C, \delta > 0$  such that every  $K \in \mathcal{S}_2(\ell^2)$  with  $\|K\|_{\mathcal{S}_2} \leq \delta$  satisfies*

$$c\|K\|_{\mathcal{S}_2} \leq h(\nu_{I+K}, \mu) \leq C\|K\|_{\mathcal{S}_2}. \quad (54)$$

*The constants can be chosen with  $\delta < 1$ , so  $I + K$  is automatically invertible. Under the weaker assumptions (50), the same conclusion holds for orthogonal  $I + K$  whenever  $f$  is non-Gaussian. This includes finite-rank perturbations whose ranges have infinite coordinate support.*

*Proof.* In the general linear case, take  $K_n = P_n K P_n$  and apply (35) to its finite coordinate block. Its untouched tail factors contribute affinity one, so the finite-dimensional Hellinger distance is also  $h(\nu_{I+K_n}, \mu)$ . The likelihood closure in Theorem 5.1 gives

$$h(\nu_{I+K_n}, \mu) = \left\| \sqrt{\frac{d\nu_{I+K_n}}{d\mu}} - 1 \right\|_2 \longrightarrow \left\| \sqrt{\frac{d\nu_{I+K}}{d\mu}} - 1 \right\|_2 = h(\nu_{I+K}, \mu).$$

Also  $\|K_n\|_{\mathcal{S}_2} \rightarrow \|K\|_{\mathcal{S}_2}$ . Passing to the limit proves both bounds.

In the orthogonal case, use the finite-coordinate polar approximants  $Q_n$  and their likelihood closure from Theorem 5.2. They satisfy  $Q_n \rightarrow I + K$  in Hilbert–Schmidt norm. Choose the radius in this assertion to be half the finite-dimensional radius in (52). Then all sufficiently large  $Q_n$  lie in that finite-dimensional chart, and

$$h(\nu_{Q_n}, \mu) \longrightarrow h(\nu_{I+K}, \mu), \quad \|Q_n - I\|_{\mathcal{S}_2} \longrightarrow \|K\|_{\mathcal{S}_2}.$$

Pass to the limit in (52). □

### 5.2.1 Characteristic-function rigidity and arbitrary mixing

Throughout this subsection,  $X_j$  are independent copies of a real random variable  $X$  with

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1,$$

and the law of  $X$  is not Gaussian. The characteristic-function arguments below require no moments of order greater than two. Write

$$\phi(t) = \mathbb{E}e^{itX}, \quad \Sigma_f := \{\varepsilon \in \{-1, 1\} : \text{Law}(\varepsilon X) = \text{Law}(X)\}.$$

Let  $\mathcal{P}_f$  be the group of coordinate permutations with row signs in  $\Sigma_f$ . When the common law has density  $f$ , this is  $\mathcal{P}_\pm$  if  $f$  is even, and the ordinary permutation group if  $f$  is not even.

**Lemma 5.4** (A product approximation for small coefficients). *There is a nonnegative function  $\eta$ , with  $\eta(\delta) \rightarrow 0$  as  $\delta \downarrow 0$ , such that, for every finite or countable real family  $(b_j)$  with  $\sum_j b_j^2 < \infty$ ,*

$$\left| \prod_j \phi(tb_j) - \exp\left(-\frac{t^2}{2} \sum_j b_j^2\right) \right| \leq \eta(\delta) t^2 \sum_j b_j^2 \quad \text{if} \quad \sup_j |tb_j| \leq \delta. \quad (55)$$

For a countable family, the product denotes the characteristic function of the  $L^2$  sum  $\sum_j b_j X_j$ .

*Proof.* The centered second-moment assumption gives

$$\phi(u) = 1 - \frac{u^2}{2} + o(u^2).$$

Consequently

$$\eta(\delta) := \sup_{0 < |u| \leq \delta} \frac{|\phi(u) - e^{-u^2/2}|}{u^2} \longrightarrow 0.$$

Both  $\phi(u)$  and  $e^{-u^2/2}$  have modulus at most one. Telescoping the difference of their finite products therefore bounds it by the sum of the differences of the factors, proving (55) for a finite family. For a countable family, pass to the limit through finite partial sums in  $L^2$  and hence in distribution. This argument does not take a logarithm of the characteristic function. □

**Lemma 5.5** (Qualitative rigidity of a unit row). *Let  $a^{(n)} \in \ell^2$  satisfy  $\|a^{(n)}\|_2 = 1$ . If*

$$\text{Law}\left(\sum_j a_j^{(n)} X_j\right) \Longrightarrow \text{Law}(X),$$

then

$$\inf_{j \geq 1, \varepsilon \in \Sigma_f} \|a^{(n)} - \varepsilon e_j\|_2 \longrightarrow 0. \quad (56)$$

*Proof.* We first describe the limits of such coefficient arrays. Permuting coefficients does not change the law of their sum. Arrange each row in nonincreasing order of absolute value, retaining its signs and padding with zeros if necessary. Its  $j$ th coefficient then has modulus at most  $j^{-1/2}$ . From any subsequence we may extract a further subsequence on which each coefficient converges, say  $a_j^{(n)} \rightarrow a_j$ . Fatou's lemma gives

$$q := \sum_j a_j^2 \leq 1.$$

For a fixed  $m$ , apply [Theorem 5.4](#) to the coefficients with  $j > m$ . For every fixed  $t$ , the error in replacing their characteristic function by

$$\exp\left[-\frac{t^2}{2} \left(1 - \sum_{j \leq m} (a_j^{(n)})^2\right)\right]$$

is at most  $t^2 \eta(|t|/\sqrt{m+1})$ , uniformly in  $n$ . Fix any  $t$ . First let  $n \rightarrow \infty$ , and then let  $m \rightarrow \infty$ . The limiting characteristic function is

$$e^{-(1-q)t^2/2} \prod_j \phi(a_j t). \quad (57)$$

Equivalently, the limiting law is that of

$$\sum_j a_j X_j + \sqrt{1-q} Z,$$

where  $Z$  is standard Gaussian and independent of the  $X_j$ .

Under the convergence hypothesis this limiting law is the law of  $X$ . Suppose that no  $|a_j|$  is one. If all  $a_j$  vanish, (57) already makes  $X$  Gaussian. Otherwise the maximum  $\rho := \max_j |a_j|$  exists, since  $(a_j) \in \ell^2$ , and satisfies  $0 < \rho < 1$ . Iterating the characteristic-function identity (57)  $k$  times gives

$$\phi(t) = e^{-(1-q^k)t^2/2} \prod_{\alpha \in \mathbb{N}^k} \phi(t b_\alpha), \quad b_\alpha = a_{\alpha_1} \cdots a_{\alpha_k}. \quad (58)$$

Here

$$\sum_\alpha b_\alpha^2 = q^k, \quad \sup_\alpha |b_\alpha| \leq \rho^k.$$

The identity may be obtained by substituting independent copies into the  $L^2$  series; at each finite depth all series have summable coefficient squares. Applying [Theorem 5.4](#) to the last product in (58) yields

$$|\phi(t) - e^{-t^2/2}| \leq t^2 q^k \eta(|t| \rho^k) \longrightarrow 0.$$

This again makes  $X$  Gaussian. This is a contradiction. Thus some  $a_j = \varepsilon \in \{-1, 1\}$ ; all the remaining coefficients and the Gaussian residual vanish. The limiting identity then says  $\text{Law}(\varepsilon X) = \text{Law}(X)$ , so  $\varepsilon \in \Sigma_f$ .

To conclude, suppose (56) failed. Take a subsequence whose distance from all the indicated unit vectors is bounded below. The preceding extraction produces a coefficient converging to an allowed sign of modulus one. Since the row norms are one, its squared distance from that signed unit vector tends to zero:

$$\|a^{(n)} - \varepsilon e_j\|_2^2 = 2 - 2\varepsilon a_j^{(n)} \longrightarrow 0.$$

Undoing the coefficient permutations gives the required contradiction.  $\square$

**Lemma 5.6** (Uniform invariance under tail symmetries). *Let  $\mu$  be the i.i.d. product law and let  $\nu \ll \mu$ . Let  $\mathcal{S}_N$  be the finite coordinate permutations, with allowed coordinate signs, that act identically on coordinates  $1, \dots, N$ . Then*

$$\sup_{S \in \mathcal{S}_N} h(\nu, S_{\#}\nu) \longrightarrow 0. \quad (59)$$

*Also, the one-coordinate marginals of  $\nu$  converge to the common coordinate law of  $\mu$ , uniformly over coordinates greater than  $N$ .*

*Proof.* Put  $p = d\nu/d\mu$  and  $p_N = \mathbb{E}_{\mu}[p \mid \sigma(x_1, \dots, x_N)]$ . The martingale convergence theorem gives  $\|p - p_N\|_{L^1(\mu)} \rightarrow 0$ . Every  $S \in \mathcal{S}_N$  preserves  $\mu$  and satisfies  $p_N \circ S^{-1} = p_N$ . Thus

$$h(\nu, S_{\#}\nu)^2 \leq \|p - p \circ S^{-1}\|_{L^1(\mu)} \leq 2\|p - p_N\|_{L^1(\mu)},$$

which proves (59). For  $i > N$ , the marginal density is  $\mathbb{E}_{\mu}[p \mid x_i]$ . Independence and  $\mathbb{E}_{\mu} p_N = 1$  imply  $\mathbb{E}_{\mu}[p_N \mid x_i] = 1$ , so

$$\|\mathbb{E}_{\mu}[p \mid x_i] - 1\|_{L^1(\mu)} \leq \|p - p_N\|_{L^1(\mu)}.$$

This proves the uniform marginal assertion.  $\square$

**Lemma 5.7** (Compact covariance rigidity from a second moment). *Let  $T$  be bounded on  $\ell^2$ . If  $T_{\#}\mu \ll \mu$ , then*

$$TT^* - I \text{ is compact.}$$

*If  $T$  is boundedly invertible,  $T^*T - I$  is compact as well.*

*Proof.* We first record uniform integrability of the squares of the linear sums

$$Z_b := \sum_j b_j X_j, \quad \|b\|_2 \leq B, \quad (60)$$

for each fixed  $B < \infty$ . Let

$$X^{(M)} = X \mathbf{1}_{\{|X| \leq M\}} - \mathbb{E}[X \mathbf{1}_{\{|X| \leq M\}}], \quad Z_b^{(M)} = \sum_j b_j X_j^{(M)}.$$

The variables  $X^{(M)}$  are bounded and centered, and converge to  $X$  in  $L^2$ . Independence gives

$$\sup_{\|b\|_2 \leq B} \|Z_b - Z_b^{(M)}\|_2 \leq B \|X - X^{(M)}\|_2 \longrightarrow 0.$$

For fixed  $M$ , expanding the fourth moment of a sum of independent centered bounded variables gives

$$\sup_{\|b\|_2 \leq B} \mathbb{E}|Z_b^{(M)}|^4 \leq C_M B^4 < \infty.$$

The bound first holds for finite sums; it also makes their truncations Cauchy in  $L^4$ , proving the countable version. Moreover,

$$\sup_{\|b\|_2 \leq B} \|Z_b^2 - (Z_b^{(M)})^2\|_1 \leq B^2 \|X - X^{(M)}\|_2 (\|X\|_2 + \|X^{(M)}\|_2) \longrightarrow 0.$$

Thus the squares in (60) are uniformly approximated in  $L^1$  by uniformly integrable families, and are themselves uniformly integrable. This uses fourth moments of bounded truncations, not a fourth-moment assumption on  $X$ .

Write  $\nu = T_{\#}\mu$ ,  $p = d\nu/d\mu$ , and  $p_N = \mathbb{E}_{\mu}[p \mid \sigma(x_1, \dots, x_N)]$ . For a finitely supported unit vector  $a$  supported on coordinates greater than  $N$ , put  $F_a(x) = (\sum_i a_i x_i)^2$ . Under  $\mu$ , these are squares from (60) with  $B = 1$ ; under  $\nu$ , they are squares of  $Z_{T^*a}$  under  $\mu$ , with  $\|T^*a\|_2 \leq \|T\|$ . They are therefore uniformly integrable under both laws. The truncated statistic  $F_a \wedge R$  depends only on tail coordinates. Independence yields

$$|\mathbb{E}_{\nu}(F_a \wedge R) - \mathbb{E}_{\mu}(F_a \wedge R)| \leq R \|p - p_N\|_1.$$

First let  $N \rightarrow \infty$  with fixed  $R$ . Then let  $R \rightarrow \infty$ . Uniform integrability gives

$$\sup_{\substack{\|a\|_2=1, \text{ } a \text{ finitely supported} \\ \text{supp } a \subset \{N+1, N+2, \dots\}}} |\langle (TT^* - I)a, a \rangle| \longrightarrow 0.$$

Since  $TT^* - I$  is self-adjoint and bounded, density of finite vectors in the tail subspace implies

$$\|(I - P_N)(TT^* - I)(I - P_N)\|_{\text{op}} \longrightarrow 0.$$

The complementary rows and columns form a finite-rank operator. Hence  $TT^* - I$  is a norm limit of finite-rank operators and is compact. For invertible  $T$ , use  $T^*T - I = T^{-1}(TT^* - I)T$ .  $\square$

**Lemma 5.8** (Tail rigidity without higher moments). *Assume that the common law has a positive density  $f$ , that  $g = \sqrt{f} \in W^{1,2}(\mathbb{R})$ , and that  $xg' \in L^2(\mathbb{R})$ . If  $T \in \text{GL}(\ell^2)$  and  $T_{\#}\mu \sim \mu$ , then*

$$T - Q \in \mathcal{S}_2 \quad \text{for some } Q \in \mathcal{P}_f. \quad (61)$$

When  $T$  is orthogonal, the condition  $xg' \in L^2$  can be omitted.

*Proof.* Write  $r_i = T^*e_i$ . By Theorem 5.7,  $\|r_i\|_2 \rightarrow 1$ . By Theorem 5.6, the law of  $\sum_j (r_i)_j X_j$  converges to the common coordinate law. Normalizing  $r_i$  to a unit vector changes this sum by a quantity tending to zero in  $L^2$ . Therefore Theorem 5.5 gives indices  $\pi(i)$  and signs  $\varepsilon_i \in \Sigma_f$ , outside a finite initial set, with

$$s_i := \varepsilon_i e_{\pi(i)}, \quad \eta_i := \|T^*e_i - s_i\|_2^2 \longrightarrow 0. \quad (62)$$

The indices  $\pi(i)$  are distinct sufficiently far out. Indeed, if  $\pi(i) = \pi(j)$  for two such rows, then  $\varepsilon_i T^*e_i - \varepsilon_j T^*e_j$  has arbitrarily small norm by (62), whereas bounded invertibility gives the lower bound  $\sqrt{2}/\|T^{-1}\|$ . Hence we can fix a cutoff after which  $\pi$  is injective.

There is also column approximation along this matching. Compactness of  $T^*T - I$  and the fact that  $\pi(i)$  leaves every finite set imply

$$\|Ts_i\|_2^2 \longrightarrow 1.$$

Since  $\langle Ts_i, e_i \rangle = \langle s_i, r_i \rangle \rightarrow 1$ , we obtain

$$\zeta_i := \|Ts_i - e_i\|_2^2 \longrightarrow 0, \quad \rho_i := \eta_i + \zeta_i \longrightarrow 0. \quad (63)$$

We prove that  $\sum_i \rho_i < \infty$  on this tail. By [Theorem 5.3](#), there are  $c_0, \delta_0 > 0$  such that

$$h(\mu, W_{\#}\mu) \geq c_0 \|W - I\|_{\mathcal{S}_2} \quad \text{when } W \in \text{GL}(\ell^2), \quad \|W - I\|_{\mathcal{S}_2} \leq \delta_0. \quad (64)$$

Suppose instead that  $\sum_i \rho_i = \infty$ . Choose  $\delta > 0$  small enough that  $3\|T^{-1}\|\delta < \delta_0$ . In any sufficiently remote tail, where  $\rho_i \leq \delta^2$ , choose a finite set  $E$  satisfying

$$\delta^2 \leq \sum_{i \in E} \rho_i \leq 2\delta^2. \quad (65)$$

Let  $q = |E|$ . Pair each  $i \in E$  with a distinct auxiliary index  $k_i$  beyond  $E$ , and set  $G = \{k_i : i \in E\}$ . Since  $\rho_i \rightarrow 0$ , these auxiliary indices may be chosen with an arbitrarily small value of  $\sum_{k \in G} \rho_k$ , while  $E$  remains fixed.

Let  $S$  exchange each output pair  $(e_i, e_{k_i})$ , and let  $P$  exchange the aligned signed input pairs  $(s_i, s_{k_i})$ , acting identically on their orthogonal complements. Both are involutions in  $\mathcal{P}_f$ , so they preserve  $\mu$ . Define

$$W = PT^{-1}ST.$$

Then

$$W - I = PT^{-1}(ST - TP), \quad \frac{\|ST - TP\|_{\mathcal{S}_2}}{\|T\|} \leq \|W - I\|_{\mathcal{S}_2} \leq \|T^{-1}\| \|ST - TP\|_{\mathcal{S}_2}. \quad (66)$$

These are finite-rank differences. Indeed  $S - I$  and  $P - I$  have finite rank. Hellinger invariance and  $P_{\#}\mu = \mu$  give

$$h(\mu, W_{\#}\mu) = h(T_{\#}\mu, S_{\#}T_{\#}\mu). \quad (67)$$

All maps in use lie in a countable operator group, so [Theorem 4.1](#) justifies this identity on a common carrier.

We estimate the commutator in (66) next. Let  $R_G$  denote the projection onto output vectors  $(e_k)_{k \in G}$ . Let  $C_G$  denote the projection onto input vectors  $(s_k)_{k \in G}$ . Set

$$J_G x = \sum_{k \in G} \langle x, s_k \rangle e_k, \quad T_0 = (I - R_G)T(I - C_G) + J_G.$$

Thus  $T_0$  has exact matched rows and columns at all the auxiliary coordinates. Orthogonal block decomposition then gives

$$\|T - T_0\|_{\mathcal{S}_2}^2 \leq \sum_{k \in G} (\|T^*e_k - s_k\|_2^2 + \|Ts_k - e_k\|_2^2) = \sum_{k \in G} \rho_k. \quad (68)$$

For completeness, the left side counts the squared entries of the auxiliary row and column errors once, while their intersection is counted twice on the right.



Order the output subspaces as  $E$ , its complement outside  $E \cup G$ , and  $G$ . Order the input subspaces as the aligned signed coordinates for  $E$ , their complement outside the aligned  $E \cup G$ , and the aligned coordinates for  $G$ . In these orthonormal decompositions,

$$T_0 = \begin{pmatrix} A & B & 0 \\ C & D & 0 \\ 0 & 0 & I_q \end{pmatrix}, \quad ST_0 - T_0P = \begin{pmatrix} 0 & -B & I_q - A \\ C & 0 & -C \\ A - I_q & B & 0 \end{pmatrix}.$$

The blocks  $B, C$  have finite-dimensional range or domain and are Hilbert–Schmidt. Put

$$\rho_E(T_0) := \sum_{i \in E} (\|T_0^* e_i - s_i\|_2^2 + \|T_0 s_i - e_i\|_2^2) = 2\|A - I_q\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2.$$

The displayed commutator gives the exact identity

$$\|ST_0 - T_0P\|_{\mathcal{S}_2}^2 = 2(\|A - I_q\|_F^2 + \|B\|_{\mathcal{S}_2}^2 + \|C\|_{\mathcal{S}_2}^2),$$

and therefore

$$\rho_E(T_0) \leq \|ST_0 - T_0P\|_{\mathcal{S}_2}^2 \leq 2\rho_E(T_0). \quad (69)$$

As the auxiliary error in (68) tends to zero,  $\rho_E(T_0) \rightarrow \sum_{i \in E} \rho_i$ . Also,

$$\|(ST - TP) - (ST_0 - T_0P)\|_{\mathcal{S}_2} \leq 2\|T - T_0\|_{\mathcal{S}_2} \rightarrow 0.$$

By (65) and (69), we can choose actual finite auxiliary indices so that

$$\frac{\delta}{2} \leq \|ST - TP\|_{\mathcal{S}_2} \leq 3\delta.$$

It follows from (66) that

$$\frac{\delta}{2\|T\|} \leq \|W - I\|_{\mathcal{S}_2} \leq 3\|T^{-1}\|\delta < \delta_0.$$

The local bound (64) now forces  $h(\mu, W_{\#}\mu) \geq c_0\delta/(2\|T\|)$ . On the other hand,  $S$  is supported in the original remote tail. [Theorem 5.6](#) makes the right side of (67) uniformly smaller than this positive number when that tail is far enough out. This is the contradiction. Therefore  $\sum_i \rho_i < \infty$ .

Define  $Q_0$  by the matched rows  $s_i$  after the fixed initial cutoff  $N_0$ , and by zero rows for  $i \leq N_0$ . The summability just proved implies

$$\|T - Q_0\|_{\mathcal{S}_2}^2 = \sum_{i \leq N_0} \|T^* e_i\|_2^2 + \sum_{i > N_0} \eta_i < \infty.$$

Therefore  $Q_0$  is a compact perturbation of an invertible operator and is Fredholm of index zero. Its cokernel consists precisely of the first  $N_0$  output coordinates. Its kernel is spanned by the unused input coordinates. There are exactly  $N_0$  unused columns. Match them to the first  $N_0$  rows with sign  $+1$ . This completes  $Q_0$  to a single  $Q \in \mathcal{P}_f$  and proves (61).

If  $T$  is orthogonal, both covariance defects vanish, and every relative commutator  $W = PT^*ST$  constructed above is orthogonal. Consequently only the orthogonal part of [Theorem 5.3](#) is needed. That part uses location Fisher information alone, so the condition  $xg' \in L^2$  may be omitted in this case.  $\square$

### 5.3 Exact identification of the linear stabilizer

The equivalence theorem needs the Sobolev assumptions. It constructs and controls likelihoods. Exact invariance is more rigid. It requires only the second moment.

**Proposition 5.9** (Exact linear stabilizer under a second-moment hypothesis). *For a centered, variance-one, non-Gaussian i.i.d. product law, the bounded invertible linear stabilizer is exactly  $\mathcal{P}_f$ . This assertion requires neither a density nor Fisher information.*

*Proof.* Every member of  $\mathcal{P}_f$  preserves the product law. Conversely, if  $T_{\#}\mu = \mu$ , covariance gives  $TT^* = I$ . Since  $T$  is boundedly invertible, it is orthogonal. Each of its unit rows gives a linear combination with the same law as  $X$ . Applying [Theorem 5.5](#) to the constant sequence consisting of this row shows that the row is an allowed signed unit vector. Orthogonality makes the selected coordinates a bijection. Thus  $T \in \mathcal{P}_f$ .  $\square$

### 5.4 Two-sided uniform integrability and completion of the proof

There are two likelihood limits in the argument. The finite-corner approximants in [Theorem 5.1](#) converge in  $L^1(\mu)$ . Their square roots converge in  $L^2(\mu)$ . Independently, once equivalence is known, the actual projected likelihoods and their reciprocals are the two conditional-expectation martingales in (42)–(43). This gives uniform integrability in both directions. It also rules out loss of mass at the projective limit.

*Proof of Theorem 1.1.* Necessity follows from [Theorem 5.8](#). If  $T - Q \in \mathcal{S}_2$  for  $Q \in \mathcal{P}_f$ , then  $TQ^{-1} = I + (T - Q)Q^{-1}$  is invertible and differs from the identity by a Hilbert–Schmidt operator. By [Theorem 5.1](#),  $\nu_{TQ^{-1}} \sim \mu$ . Since  $Q_{\#}\mu = \mu$ , this is  $\nu_T \sim \mu$ . The same proposition gives the finite-corner likelihood and square-root limits stated in the theorem. The exact stabilizer is identified in [Theorem 5.9](#). Finally, [Theorems 4.2](#) and [4.3](#) give the projected-affinity characterization, the singular alternative, and both uniform-integrability statements.  $\square$

**Theorem 5.10** (Complete orthogonal criterion with only location Fisher information). *Let  $f > 0$  almost everywhere be a centered, variance-one, non-Gaussian density, and assume only  $\sqrt{f} \in W^{1,2}(\mathbb{R})$  in addition to its second moment. For every orthogonal  $U$  on  $\ell^2$ ,*

$$U_{\#}\mu \sim \mu \iff U - Q \in \mathcal{S}_2 \text{ for some } Q \in \mathcal{P}_f.$$

*If this condition fails,  $U_{\#}\mu \perp \mu$ .*

*Proof.* Necessity is [Theorem 5.8](#). For sufficiency, write  $UQ^{-1} = I + (U - Q)Q^{-1}$ . This is orthogonal and differs from the identity by a Hilbert–Schmidt operator. Apply [Theorem 5.2](#), and then use  $Q_{\#}\mu = \mu$ . The alternative follows from [Theorem 4.3](#).  $\square$

### 5.5 Globalization of finite-dimensional coercivity

We finish the proof deferred in [Section 3.4](#). The key point is this. Failure of a dimension-free separation estimate would create a single forbidden infinite-dimensional equivalence.

*Proof of Theorem 3.6.* Fix  $0 < a \leq b < \infty$ . We first prove uniform separation away from the stabilizer: for every  $\varepsilon > 0$ ,

$$\alpha_{\varepsilon,a,b} := \inf \left\{ h(\mu_n, A_{\#}\mu_n)^2 : \begin{array}{l} n \geq 1, A \in \text{GL}(n, \mathbb{R}), \\ s_{\min}(A) \geq a, s_{\max}(A) \leq b, \\ d_{n,f}(A) \geq \varepsilon \end{array} \right\} > 0. \quad (70)$$

Suppose otherwise. Then there are dimensions  $n_k$  and matrices  $A_k$  in the displayed class with  $h(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) \rightarrow 0$ . Since

$$-2 \log A(\mu_{n_k}, (A_k)_{\#}\mu_{n_k}) = -2 \log \left( 1 - \frac{1}{2} h(\mu_{n_k}, (A_k)_{\#}\mu_{n_k})^2 \right),$$

we may pass to a subsequence. Without changing notation, assume this sum of block energies is finite.

Decompose  $H = \bigoplus_{k \geq 1} \mathbb{R}^{n_k}$  into consecutive coordinate blocks. Now define  $A = \bigoplus_{k \geq 1} A_k$ . The common singular-value bounds imply  $A \in \text{GL}(H)$ . Hellinger affinity tensorizes across the blocks, so the exact block-product criterion ?? gives  $A_{\#}\mu \sim \mu$ . The already proved [Theorem 1.1](#) therefore supplies  $Q \in \mathcal{P}_f$  such that

$$D := A - Q \in \mathcal{S}_2(H). \quad (71)$$

Let  $R_k$  be the orthogonal projection onto the  $k$ -th output block. The orthogonality of these blocks gives

$$\sum_k \|R_k D\|_{\mathcal{S}_2}^2 = \|D\|_{\mathcal{S}_2}^2 < \infty, \quad \text{hence} \quad \|R_k D\|_{\mathcal{S}_2} \rightarrow 0.$$

If, for some output basis vector  $e_i$  in the  $k$ -th block, the signed basis vector  $Q^* e_i$  lies outside the corresponding input block, then  $A^* e_i$  and  $Q^* e_i$  have disjoint supports. The lower singular value bound then yields

$$\|A^* e_i - Q^* e_i\|_2^2 = \|A^* e_i\|_2^2 + 1 \geq a^2 + 1. \quad (72)$$

For all sufficiently large  $k$ , this contradicts  $\|R_k D\|_{\mathcal{S}_2}^2 < a^2 + 1$ . Hence  $Q^*$  maps every output coordinate of each sufficiently large block into the same input block. Its restriction is injective between two sets of cardinality  $n_k$ . Hence it is a bijection. Denote the resulting member of  $\mathcal{P}_f(n_k)$  by  $Q_k$ . It follows that

$$d_{n_k, f}(A_k) \leq \|A_k - Q_k\|_F = \|R_k D\|_{\mathcal{S}_2} \rightarrow 0,$$

contradicting  $d_{n_k, f}(A_k) \geq \varepsilon$ . This proves (70).

Let  $c_0, C_0, \delta_0$  be the constants in [Theorem 3.5](#), decreasing  $\delta_0$  so that  $0 < \delta_0 < 1$ . Given  $T$ , choose a minimizer  $Q \in \mathcal{P}_f(n)$  in (36). Since  $Q_{\#}\mu_n = \mu_n$ ,

$$h(\mu_n, T_{\#}\mu_n) = h(\mu_n, (TQ^{-1})_{\#}\mu_n), \quad \|TQ^{-1} - I\|_F = d_{n, f}(T).$$

When  $d_{n, f}(T) \leq \delta_0$ , the local theorem supplies the two-sided linear bounds for  $h$ . When  $d_{n, f}(T) > \delta_0$ , (70) with  $\varepsilon = \delta_0$  supplies the lower bound, while  $h^2 \leq 2$  supplies the upper bound. More explicitly, one may take

$$c = \min\{c_0^2, \alpha_{\delta_0, a, b}\}, \quad C = \max\{C_0^2, 2\delta_0^{-2}\}.$$

These constants give (37) in all regimes. □

## 6 Scope and Sharpness of the Hypotheses

The hypotheses in (3) include centered, variance-normalized Laplace densities, despite their cusp at the origin. They also include variance-normalized densities proportional to  $e^{-x^4}$ . Symmetry is not needed in any step of the proof.

For a family allowing both asymmetry and heavy tails, take

$$\tilde{f}_{\nu,\beta}(y) = c_{\nu,\beta}(1+y^2)^{-(\nu+1)/2}e^{\beta \arctan y}, \quad \nu > 2, \quad \beta \in \mathbb{R}, \quad (73)$$

and center and rescale to variance one. The unstandardized location score is

$$\tilde{s}(y) = \frac{(\nu+1)y - \beta}{1+y^2}.$$

Both  $\tilde{s}$  and  $y\tilde{s}$  are bounded. The density is positive and smooth. It has a finite second moment. Centering and rescaling preserve finiteness of the location and scale information. So the standardized density satisfies (3). For  $2 < \nu \leq 4$ , its fourth moment is infinite. When  $\beta \neq 0$ , the two tails have unequal leading constants. So centering does not make the density even. The case  $\beta = 0$  is a rescaled Student family. These examples show that neither evenness nor a fourth moment is implicit in the theorem.

The Gamma examples used to illustrate the Fourier response belong to Theorem 2.1. Their centered densities still vanish on a half-line. So they do not satisfy the full-support hypothesis of Theorem 1.1. A nonzero Fourier response establishes local detection. Alone, this does not establish the general measure-equivalence theorem for a density with restricted support.

The second moment has a specific role beyond differentiating one Fourier feature. It makes every  $\ell^2$  row into a convergent random series. It supplies uniform integrability of squares of linear forms. It gives the Gaussian-residual compactness argument. So the first-moment ODE result is stronger in its moment scope than the arbitrary-operator equivalence theorem. Extending the latter to infinite-variance coordinates requires a different domain of admissible rows and a corresponding operator criterion. For example, for symmetric nondegenerate  $\alpha$ -stable coordinates,  $0 < \alpha < 2$ , convergence of a row series requires  $\sum_j |a_j|^\alpha < \infty$ : its partial-sum characteristic functions are

$$\exp \left( -c|t|^\alpha \sum_{j \leq n} |a_j|^\alpha \right).$$

An arbitrary  $\ell^2$  row need not meet that requirement. The condition on admissible rows must also be satisfied when using bounded or fractional Fourier features.

The Hellinger path estimates give  $L^1$  convergence and uniform integrability of likelihoods. The finite-frequency functions are bounded in each coordinate. Their dimension-independent bounds are the  $L^2(\mu_n)$  and generator estimates proved above. No variance bound for their aggregate under an arbitrary mixed output law is needed in the tail-symmetry argument.

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